Exploration of Stochastic Processes with Computational Implementation

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Contents

1 Introduction **2**

2 Random Walks **3**
2.1 Mathematical Definition . . . 3
2.2 Theoretical Properties . . . 3
2.3 Computational Implementation . . . 4
2.4 Numerical Experiments . . . 4

3 Gambler's Ruin **6**
3.1 Problem Formulation . . . 6
3.2 Analytical Solution . . . 6
3.3 Simulating Varying Numbers of Paths to Test Convergence . . . 6
3.4 Theory vs Simulation . . . 8
3.5 How Probability of Ruin Changes With Start Position a and Probability p . . . 9
3.6 Summary . . . 10

4 Markov Chains **11**
4.1 Transition Probabilities and Matrices . . . 11
4.2 Computational Implementation and Convergence . . . 12
4.3 Gambler's Ruin as a Markov Chain . . . 13

5 Poisson processes **14**
5.1 Defining a Poisson Process . . . 14
5.2 Computational Verification . . . 14

6 Brownian Motion **16**
6.1 Definitions and Properties . . . 16
6.2 Simulating and Validating Properties . . . 17
6.3 Two Boundary Hitting Times for Brownian Motion . . . 18

7 Future Research **21**

1 Introduction

This report explores a number of stochastic processes, from random walks to Brownian motion, and pairs each analytical result with a Monte Carlo simulation in Python. For each process I derive the relevant theory, implement it computationally, and test whether the simulations agree with the theory and how quickly they converge.

The report begins with simple random walks, deriving their expectation and variance and examining how sensitive each estimate is to sample size. These ideas are then applied to the gambler's ruin problem, where I solve for the probability of ruin and expected duration using difference equations and use simulation to show that the expected duration alone can be a misleading summary of the game. Markov chains are then introduced, providing an explicit matrix method that recovers the full distribution of the game's duration and a convergence to a stationary distribution. Next, Poisson processes extend the setting to continuous time, and I verify their distributional and superposition properties. Finally, I study Brownian motion, simulating it as a Gaussian random walk, relating the spreading of its density to the diffusion equation, and solving the continuous gambler's ruin problem using martingales and optional stopping.

A recurring theme is the comparison between exact results and simulated ones, in particular the $1/\sqrt{n}$ rate at which Monte Carlo error falls. The report ends with possible extensions, including applications to financial problems.

2 Random Walks

2.1 Mathematical Definition

A random walk is a stochastic process constructed from a sequence of random increments. We would define a one-dimensional simple random walk process to go up by 1 with a probability of p and down by 1 with a probability of q. This can be written formally as,

$$X_{n+1} = \begin{cases} X_n + 1 & \text{with probability } p, \\ X_n - 1 & \text{with probability } q. \end{cases}$$

Or alternatively,

$$X_n = X_0 + \sum_{i=1}^{n} Z_i, \quad \text{where} \quad Z_i = \begin{cases} 1 & \text{with probability } p, \\ -1 & \text{with probability } q. \end{cases}$$

.

2.2 Theoretical Properties

The expectation of a random walk is defined as

$$E[X_n] = X_0 + n(p - q).$$

This can be proved using the second definition for a random walk by taking the expectation of X_n

$$E[X_n] = E[X_0] + E\left[\sum_{i=1}^{n} Z_i\right] \Rightarrow E[X_n] = X_0 + \sum_{i=1}^{n} E[Z_i]$$

and since $\mathbb{P}(Z_i = 1) = p$ and $\mathbb{P}(Z_i = -1) = q$ we have,

$$E[Z_i] = p - q$$

giving

$$E[X_n] = X_0 + \sum_{i=1}^{n} (p - q) = X_0 + n(p - q).$$

The variance of a random walk is defined as

$$Var[X_n] = 4npq.$$

Since $Var[X_0] = 0$, this can be proved simply by taking the variance of X_n

$$Var[X_n] = Var[X_0] + \sum_{i=1}^{n} Var[Z_i] = nVar[Z_1]$$

and with $E[Z_1^2] = p + q$

$$\begin{aligned} Var[Z_1] &= E[Z_1^2] - E[Z_1]^2, \\ &= (p + q) - (p - q)^2, \\ &= (p + q) - (p^2 - 2pq + q^2), \\ &= p(1 - p) + q(1 - q) + 2pq, \\ &= pq + pq + 2pq = 4pq, \\ Var[X_n] &= nVar[Z_1] = 4npq. \end{aligned}$$

2.3 Computational Implementation

The random walks were simulated by generating independent Bernoulli trials at each step, with a probability p of going up by 1 and $1-p$ going down by 1. The results from these Bernoulli trials were cumulatively summed to create the path. The simulation can be repeated for a large number of independent paths, allowing us to estimate the expectation and variance and compare them with the theoretical results.

2.4 Numerical Experiments

Figure (1) gives a visual realization to a random walk with probability $p = 0.5$ over 300 steps. Although each increment is restricted to ± 1, we observe a lot of variation over time.

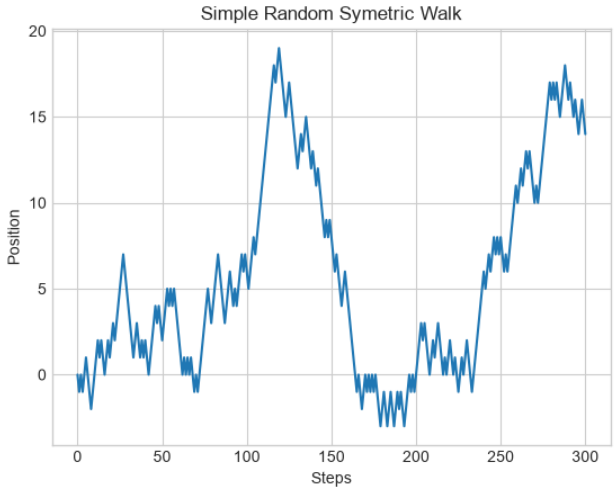


Figure 1: A single simple symmetric random walk

We next investigate many simulations with probability $p = 0.6$ over 300 steps, to test the empirical values for expectation and variance and observe the distribution of final positions produced by simulating many paths.

From the histograms in Figure (2) we can see that as n increases the distribution of final positions approaches a normal distribution. Table (1) shows the means and variances of each simulation size n.

Simulations	Mean	Mean Error	Variance	Variance Error
20	61.8	1.8	370.76	82.76
100	62.4	2.4	271.66	16.34
10000	60.1	0.1	287.30	0.7

Table 1: Table showing estimated mean and variance of final position for n simulations

Using the equations for variance and expectation we obtain theoretical values:

$$E[X_{300}] = 0 + 300 \times (0.6 - 0.4) = 60$$
$$Var[X_{300}] = 4 \times 300 \times 0.6 \times 0.4 = 288$$

We can see that as n increases the estimated values for expectation and variance become more accurate, interestingly our estimations for expectation were all pretty close to the theoretical value even with small n whilst the estimated variance was a lot less accurate. This is because the mean is estimated by averaging outcomes directly whereas variance is estimated by averaging the squared deviation from the mean. Squaring amplifies the influence of any walk that travels far from the mean and explains why for our $n = 20$ simulation the estimated variance is much larger than the theoretical since it contained a relatively extreme final position.

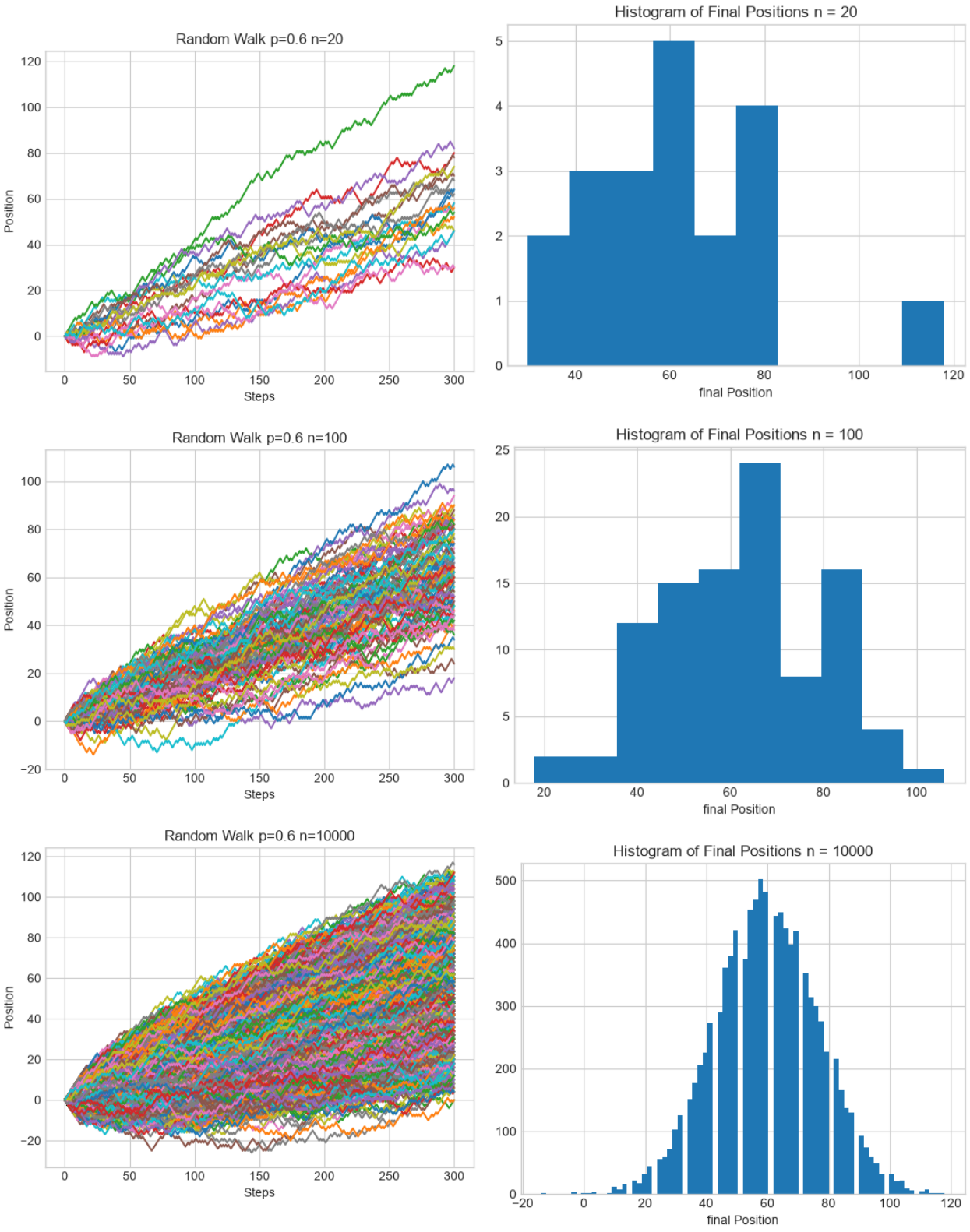


Figure 2: Simulation of many walks including histograms of final positions

3 Gambler's Ruin

3.1 Problem Formulation

Player A is playing against Player B, Player A starts with £a and Player B starts with £b where $m = a+b$ is the total amount of money. At each step in the game each player bets £1 where Player A wins £1 with probability p and loses with probability q. The game continues until either player goes bankrupt. We can represent this as a stochastic process

$$X_{n+1} = \begin{cases} X_n + 1 & \text{with probability } p \text{ if } 1 \leq X_n \leq m-1, \\ X_n - 1 & \text{with probability } q \text{ if } 1 \leq X_n \leq m-1, \\ 0 & \text{if } X_n = 0, \\ m & \text{if } X_n = m. \end{cases}$$

This is the same as a random walk but with absorbing barriers at $X_n = 0$ and $X_n = m$.

3.2 Analytical Solution

First, let's solve for the probability of ruin, that is the probability of player A going bankrupt. We can write

$$\begin{aligned} \mathbb{P}(\text{Ruin}) &= \mathbb{P}(\text{win first round})\mathbb{P}(\text{Ruin} \mid \text{win first round}) + \mathbb{P}(\text{lose first round})\mathbb{P}(\text{Ruin} \mid \text{lose first round}) \\ &= p\mathbb{P}(\text{Ruin} \mid \text{win first round}) + q\mathbb{P}(\text{Ruin} \mid \text{lose first round}) \end{aligned}$$

From the Markov property, we know that $\mathbb{P}(\text{Ruin} \mid \text{lose first round})$ and $\mathbb{P}(\text{Ruin} \mid \text{win first round})$ are the same as $\mathbb{P}(\text{Ruin})$ just with ± 1 starting points, allowing us to write the probability of ruin as

$$r_i = pr_{i+1} + qr_{i-1}.$$

giving the linear difference,

$$pr_{i+1} - r_i + qr_{i-1} = 0 \qquad \text{subject to} \qquad r_0 = 1,\, r_m = 0.$$

by solving the difference equation with a as the start point and $\rho = q/p$, the solution for the probability of ruin is

$$r_a = \begin{cases} \frac{\rho^a - \rho^m}{1-\rho^m} & \text{if } \rho \neq 1, \\ 1 - \frac{a}{m} & \text{if } \rho = 1. \end{cases}$$

Similarly, we can solve for the expected duration of game,

$$\begin{aligned} E[\text{duration}] &= \mathbb{P}(\text{win first round})E[\text{duration} \mid \text{win first round}] + \mathbb{P}(\text{lose first round})E[\text{duration} \mid \text{lose first round}] \\ &= pE[\text{duration} \mid \text{win first round}] + qE[\text{duration} \mid \text{lose first round}] \\ d_i &= p(1 + d_{i+1}) + q(1 + d_{i-1}) \end{aligned}$$

The important difference is that $E[\text{duration} \mid \text{win first round}] = 1 + d_{i+1}$ because the round itself takes 1 time step. Again we can rearrange to form the difference equation

$$pd_{i+1} - d_i + qd_{i-1} = -1 \quad \text{subject to} \quad d_0 = 0,\, d_m = 0.$$

and solving this inhomogeneous difference equation we obtain

$$d_a = \begin{cases} \frac{1}{q-p}(a - m\frac{1-\rho^a}{1-\rho^m}) & \text{if } \rho \neq 1, \\ a(m-a) & \text{if } \rho = 1. \end{cases}$$

3.3 Simulating Varying Numbers of Paths to Test Convergence

Figure (3) shows random walks simulated with probability $p = 0.5$ of a win at each step and a starting point at $a = 20$. This simulation has player B also start with £20 so $m = 40$, therefore the problem has 2 absorbing boundaries at 0 and 40 both at equal distance from the origin.

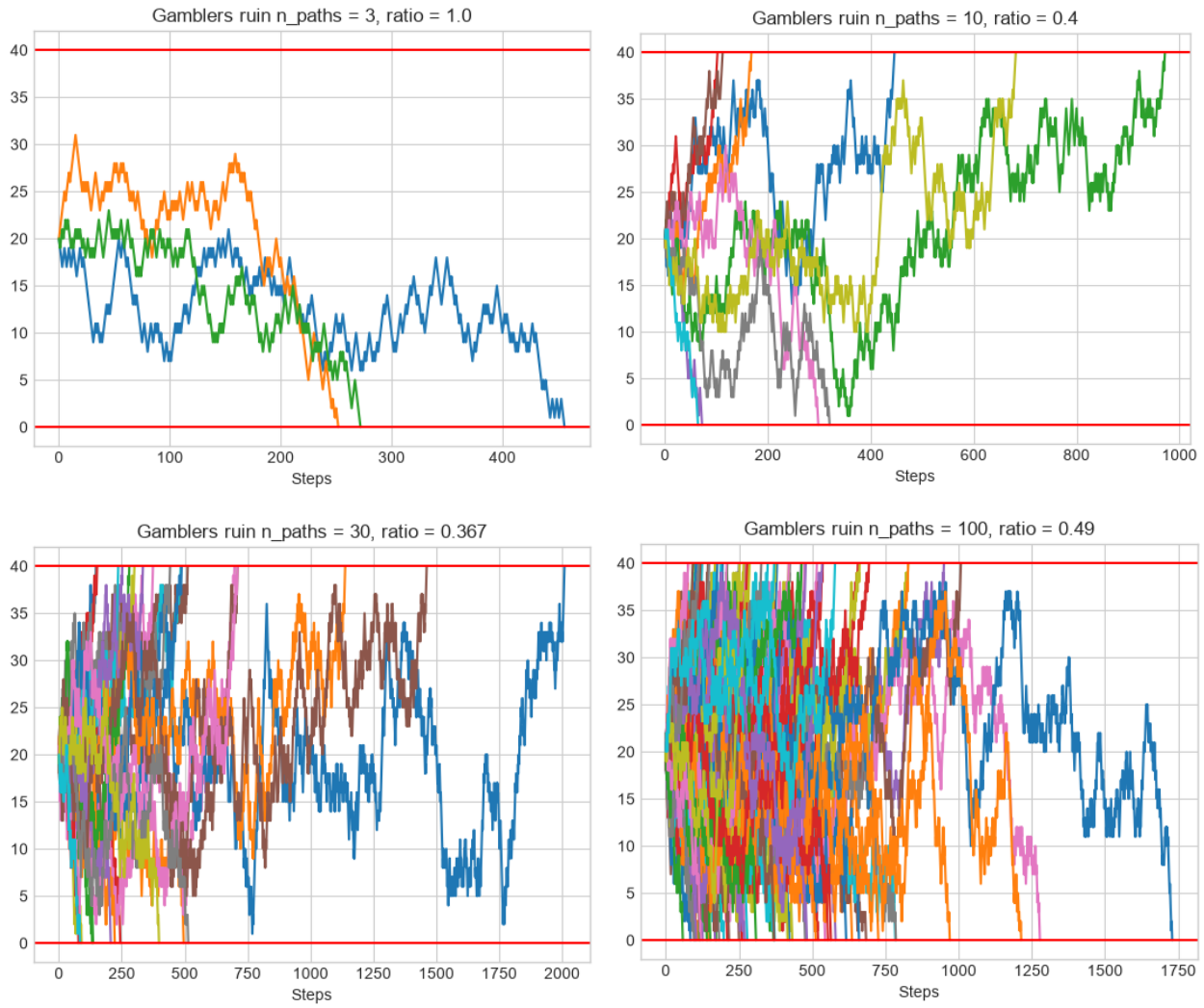


Figure 3: Simulation of gamblers ruin problem with varying numbers of paths

Without calculating the probability of ruin we can clearly see that the probability should be 0.50, and from this figure we can see that as n increases the ratio of paths that are ruined become closer to the theoretical probability.

We will investigate this further by looking at how this ratio of ruin converges as n increases, however using $m = 50$ and $p = 0.51$ and keeping $a = 20$. This time the theoretical probability isn't obvious so we will have to solve the probability of ruin using the equation stated in the last section. With $\rho = 49/51$ we have

$$r_{20} = \frac{\frac{49}{51}^{20} - \frac{49}{51}^{50}}{1 - \frac{49}{51}^{50}} = 0.3631.$$

Figure (4) includes 2 plots, the plot on the left shows how the ratio of paths that become ruined changes with an increase in n and the figure on the right shows absolute error. At first this result might not be what you expected as after the first few thousand paths we don't really observe a convergence, instead it seems to look like a white noise process.

This can be explained since we can treat each path as an independent Bernoulli trial (ruined, not ruined), with variance $r(1-r)$, where r is the theoretical probability of ruin. Since our simulated ratio $\hat{r}$ is the average of n trials, its variance is $r(1-r)/n$ and therefore the standard error is $SE[\hat{r}] = \sqrt{r(1-r)/n}$. The standard error is proportional to $1/\sqrt{n}$, which is why increasing n from 2000 to 10000, a fivefold increase, only produces a factor of $1/\sqrt{5} \approx 0.45$ reduction in the standard error.

In the figure on the right I included a plot of standard error since it gives the theoretical scale around which the error should fluctuate.

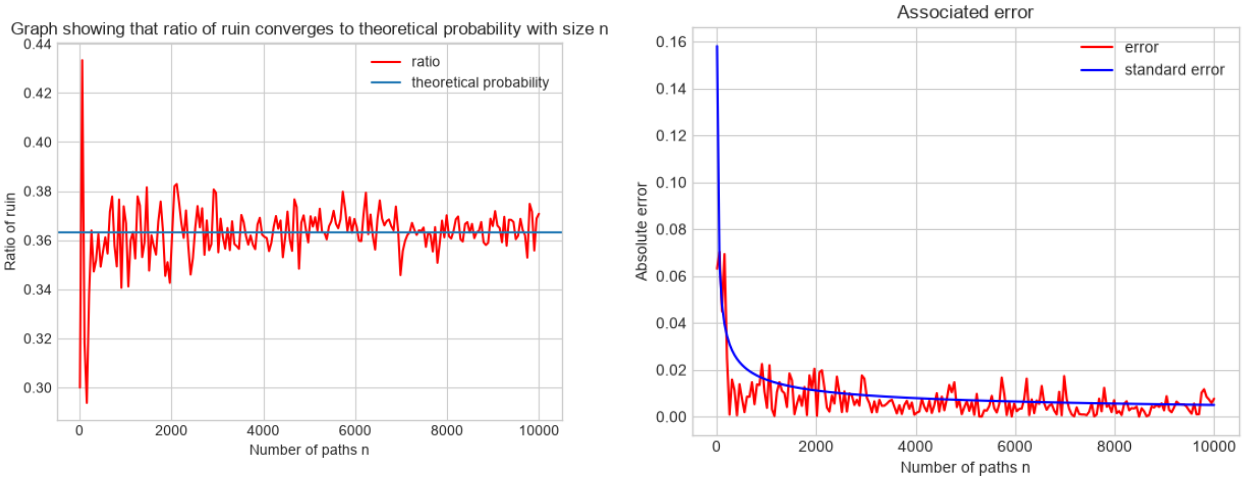


Figure 4: Simulating error with change in number of paths

3.4 Theory vs Simulation

We have looked at how probability of ruin can be calculated and simulated, let's next investigate solving duration. Using the same variables $m = 50$, $p = 0.51$ and $a = 20$ as last time we can calculate the expected duration

$$d_{20} = \frac{1}{0.49 - 0.51}(20 - 50 \times \frac{1 - \frac{49}{51}^{20}}{1 - \frac{49}{51}^{50}}) = 592.2.$$

Now we can calculate the mean duration from simulating $100,000$ paths, Figure (5) shows the mean duration and compares it with the theoretical expected duration. More importantly it shows the shape of its distribution.

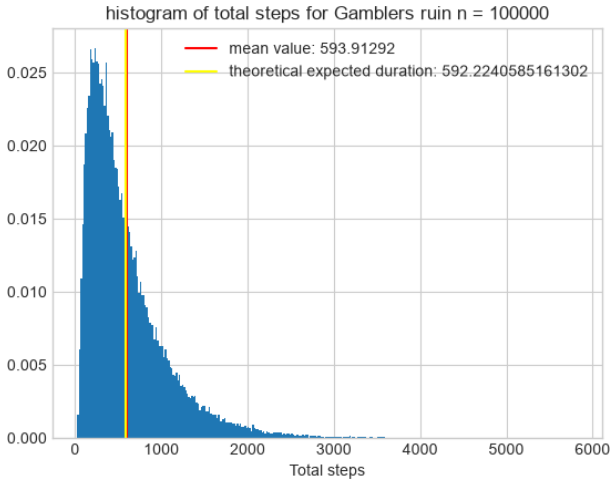


Figure 5: Histogram of total steps

Whilst having the equation to work out theoretical expected duration is handy especially as it matches the simulated value, in practice it would be much better to have the distribution associated with duration. That being said this distribution doesn't follow any standard binomial, poisson, normal or gamma distributions. Instead we can use python to track the probability of being at each position over time, splitting each step by p and q until it records how much land on 0 and m at each step. This gives the exact PMF for duration of the game shown in Figure (6).

The simulated paths appear to reinforce our method for calculating the exact PMF. This result is important as we can now find exact probabilities using this distribution rather than relying only on the expected duration or probabilities from the simulated distribution. For example, we can find the number of steps by which there is a 95% probability that the game has finished. Using a search in Python, we find that this occurs at 1521 steps.

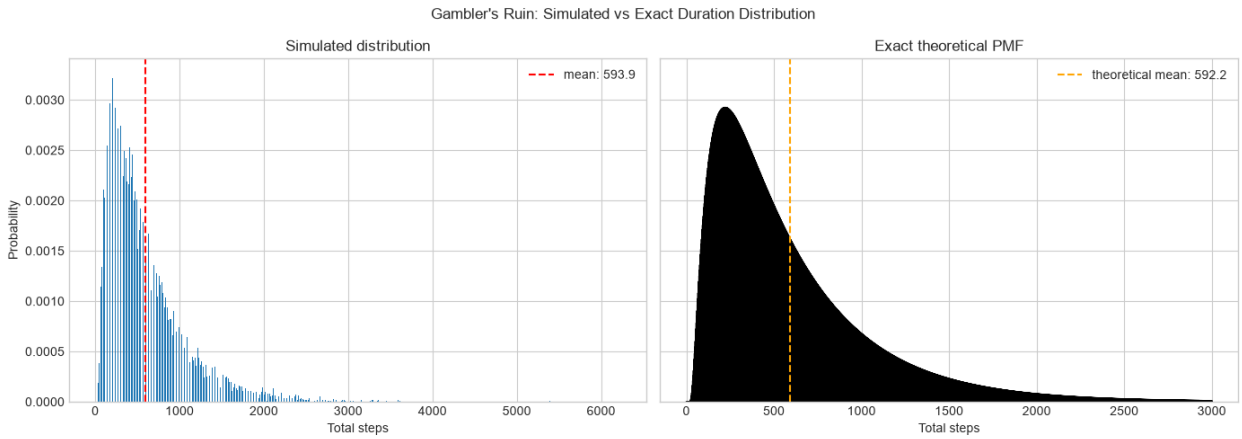


Figure 6: Simulated distribution vs probability mass function

Experimenting with values for a, m and p I was able create the PMF in Figure (7). This process is quite interesting as there is a relatively high probability of the game finishing early on as the start position of 10 is very close to 0, however because of the probability $p = 0.525$ there is a drift that means the theoretical probability of ruin is only 0.448.

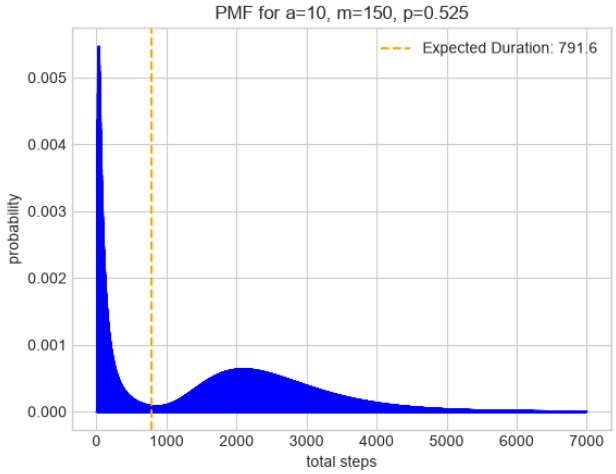


Figure 7: PMF showing the flaw in solely using expected duration

The main flaw I wanted to point out was that the expected duration alone provides an incomplete description of the distribution. In this example, the mean duration lies in a relatively low-probability region and although the average total steps would certainly match the expected duration there is only a 3% chance a game would finish in the range $[500, 1000]$ whilst there is a 33% chance finishing before and 64% chance finishing after. This demonstrates that the expected value does not necessarily correspond to a typical outcome, particularly for asymmetric distributions.

3.5 How Probability of Ruin Changes With Start Position a and Probability p

From the last few simulations or just thinking logically it follows intuitively that the closer the starting position a is to 0 and the smaller the value of p the higher the probability of ruin. Figure (8) shows that the computed ratios fit the probability of ruin for both change in start position and probability. The left plot shows a decrease in probability with increase in start position with a curved increasing gradient. From the equation for probability of ruin we can see that the equation of this curve is

$$r_a = 1.156 \times \frac{49}{51}^a - 0.156.$$

This equation describes an exponential decay, which explains the shape of the plot. If we were to change p to be less than 0.5 then with $\rho > 1$ we would expect the opposite, a decreasing concave exponential

decay.

Now focusing on the equation on the right it looks like an inverted tanh curve or s shape. In this case looking at the equation won't really tell us anything.

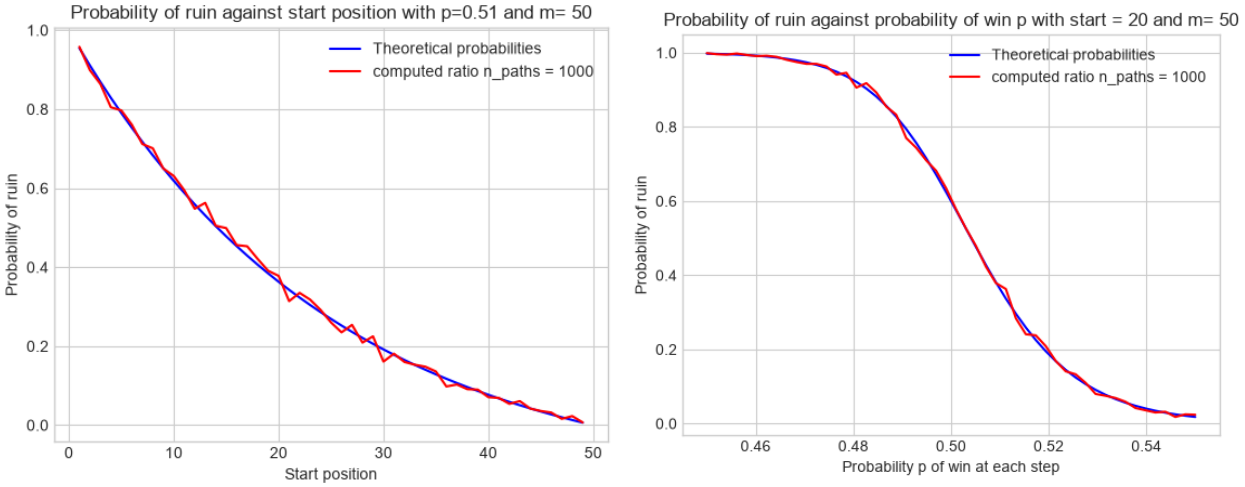


Figure 8: Probability of ruin against start position and probability of win

Instead Figure (9) shows us that an increase in a pivots the shape of the curve. Taking another look at the left graph we can see that a change in p affects the change in gradient of the curve such that curves with $p > 0.5$ are symmetric with curves with $p < 0.5$ about the curve of $p = 0.5$ where the gradient is constant.

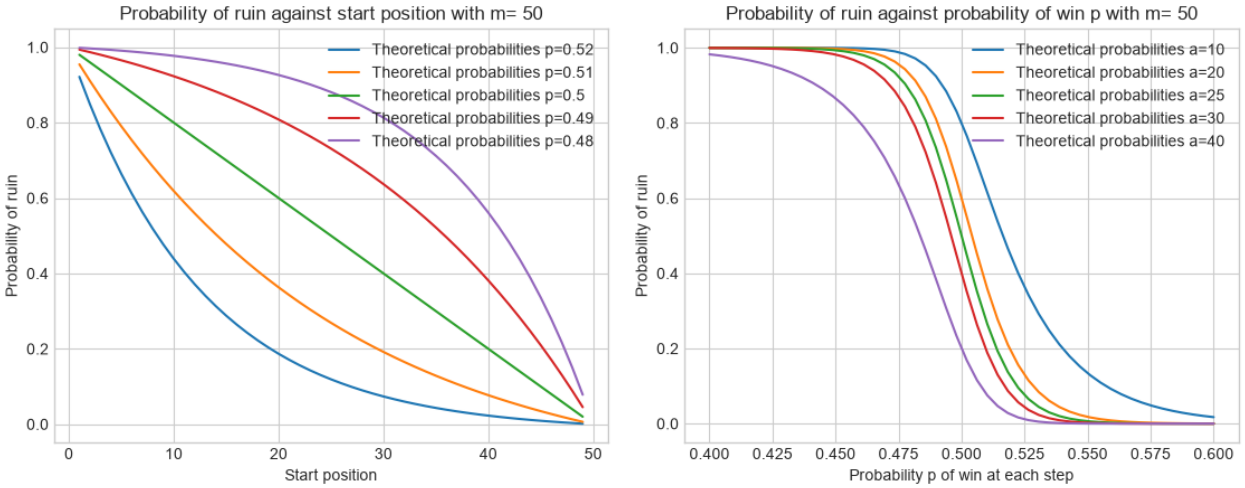


Figure 9: Probability of ruin against start position and probability of win with different variations

3.6 Summary

The Gambler's Ruin problem demonstrated how analytical solutions for both absorption probabilities and expected duration can be validated computationally. The simulations reproduced the theoretical ruin probability and expected duration, while the analysis of the duration distribution demonstrated limitations of relying solely on expected values. The problem also provided an introduction to absorbing stochastic processes and motivated the use of transition probabilities considered in the following section on Markov chains.

4 Markov Chains

4.1 Transition Probabilities and Matrices

A Markov chain is a sequence of random variables $X_0, X_1, X_2, ...$ taking values from a state space S whilst satisfying the Markov property. Which is formally written

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, ..., X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Or in words, given the present state the future is independent of the past. Recall that we used this property in solving gambler's ruin, however we used it to condition the problem on its first step and then solving the difference equation. Whilst this method works cleanly for the gambler's ruin problem, using transition probabilities/matrices gives an explicit method that works with any transition structure or number of states.

In the previous examples we also only considered a specific start point, but we could pick a start point according to some distribution $\lambda_i = \mathbb{P}(X_0 = i)$. Now consider Figure (10) a 3-state Markov chain with state space $S = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 0 & 0.3 & 0.7 \\ 0.4 & 0 & 0.6 \\ 0.5 & 0.5 & 0 \end{pmatrix}.$$

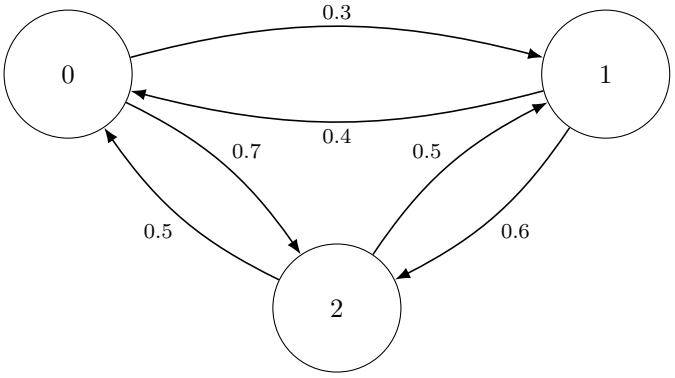


Figure 10: three state transition diagram

To find the probability of landing on a state j after n steps we can write

$$p_{ij}(n) = \mathbb{P}(X_n = j \mid X_0 = i).$$

Where $p_{ij}(n)$ is the $(i, j)th$ entry of the matrix P^n. So if we wanted to find the probability of being in state 1, three steps after starting in state 2 we would only need to calculate the matrix P^3 which can be calculated computationally. With

$$P^3 = \begin{pmatrix} 0.23 & 0.231 & 0.539 \\ 0.308 & 0.23 & 0.462 \\ 0.385 & 0.385 & 0.23 \end{pmatrix}$$

looking at the $(2, 1)$ entry we can see that the probability would be 0.385. However when considering a start distribution λ_i we instead say that the probability $\mathbb{P}(X_n = j)$ is the jth entry in the row vector of probabilities given by λP^n. So if we instead considered the start distribution of

$$\lambda_i = \mathbb{P}(X_0 = i) = \begin{cases} 0.2 & \text{if } i = 0 \\ 0.4 & \text{if } i = 1 \\ 0.4 & \text{if } i = 2 \end{cases} \qquad \lambda = \begin{pmatrix} 0.2 & 0.4 & 0.4 \end{pmatrix}$$

we now have

$$\lambda P^3 = \begin{pmatrix} 0.3232 & 0.2922 & 0.3846 \end{pmatrix}.$$

So if we still wanted to know the probability of ending on 1 whilst knowing only the start distribution, we get a probability of 0.2922.

4.2 Computational Implementation and Convergence

By simulating paths that follow the transition diagram in Figure (10) we can check if the probability calculated by λP^n is accurate to the ratio of final positions for each state after a certain number of steps. Figure (11) shows a plot for the ratio of paths with a final position of 0, 1 and 2 after n $= 20$ steps.

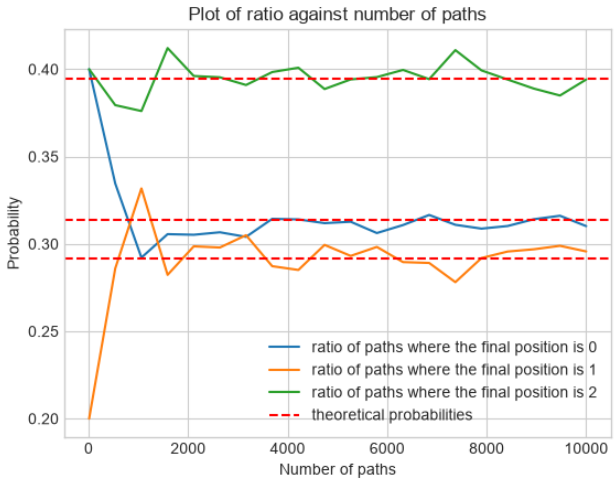


Figure 11: Plot showing the ratio of final positions for each state with increase in number of paths n

With a row vector of probabilities of

$$\lambda P^{20} = \begin{pmatrix} 0.31389876 & 0.29147617 & 0.39462507 \end{pmatrix}$$

we can see that as the number of paths increases the ratio of paths in each respective state converges closer to the theoretical value given by the row vector.

Now let's investigate how the theoretical probability of this three state Markov chain behaves with an increasing number of steps. Figure (12) shows us that the theoretical probability fluctuates for each state until around step 10 where each state seems to converge.

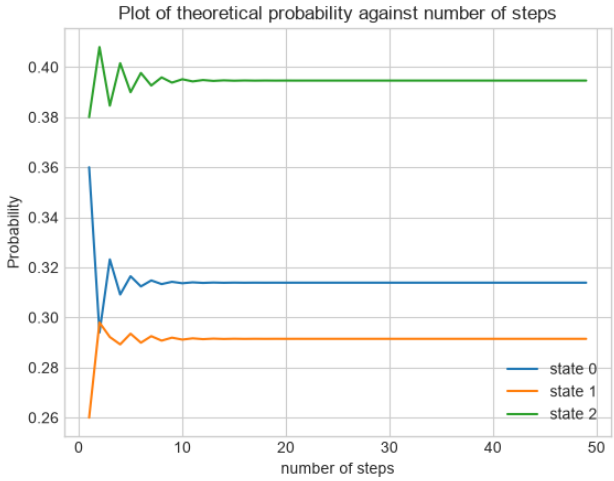


Figure 12: Figure of theoretical probability against number of steps

This happens because our Markov chain is both irreducible (every state can be reached from every other state) and aperiodic (the chain does not cycle in a fixed pattern). These properties guarantee the existence of a unique stationary distribution π satisfying $\pi P = \pi$, and the state probabilities converge to π from any initial distribution. The plot shows that no matter where we start the probability will always converge to π.

At $n = 20$ the distribution appears to have numerically converged using the row vector calculated earlier, let's verify that it approximately satisfies $\pi P = \pi$.

$$\pi P = \begin{pmatrix} 0.31389876 & 0.29147617 & 0.39462507 \end{pmatrix} \begin{pmatrix} 0 & 0.3 & 0.7 \\ 0.4 & 0 & 0.6 \\ 0.5 & 0.5 & 0 \end{pmatrix}$$
$$= \begin{pmatrix} 0.313903 & 0.29148216 & 0.39461484 \end{pmatrix} \approx \pi$$

4.3 Gambler's Ruin as a Markov Chain

Let's revise the gambler's ruin problem using the same values $m = 50$, $p = 0.51$ and $a = 20$ but instead explicitly solving using Markov chains instead of using the formulas derived by solving difference equations. We can do this using a 51×51 transition matrix where for every ith row entry $(i, i-1) = 1-p$ and $(i, i+1) = p$ apart from row 0 and m which have entries $(0, 0) = 1$ and $(m, m) = 1$ respectively. Which will look like this:

$$P = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0.49 & 0 & 0.51 & \cdots & 0 & 0 & 0 \\ 0 & 0.49 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0.51 & 0 \\ 0 & 0 & 0 & \cdots & 0.49 & 0 & 0.51 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix}$$

Using the transition matrix we can calculate the probability of being in any state at any step in time, which is particularly useful for finding the probability of ruin at any step you want. The left plot in Figure (13) was quickly computed by calculating the probability of being in the ruined state at every step until 5000. As you can see in the plot it converges at 0.3631 which is the probability of ruin calculated using the equation from the last section.

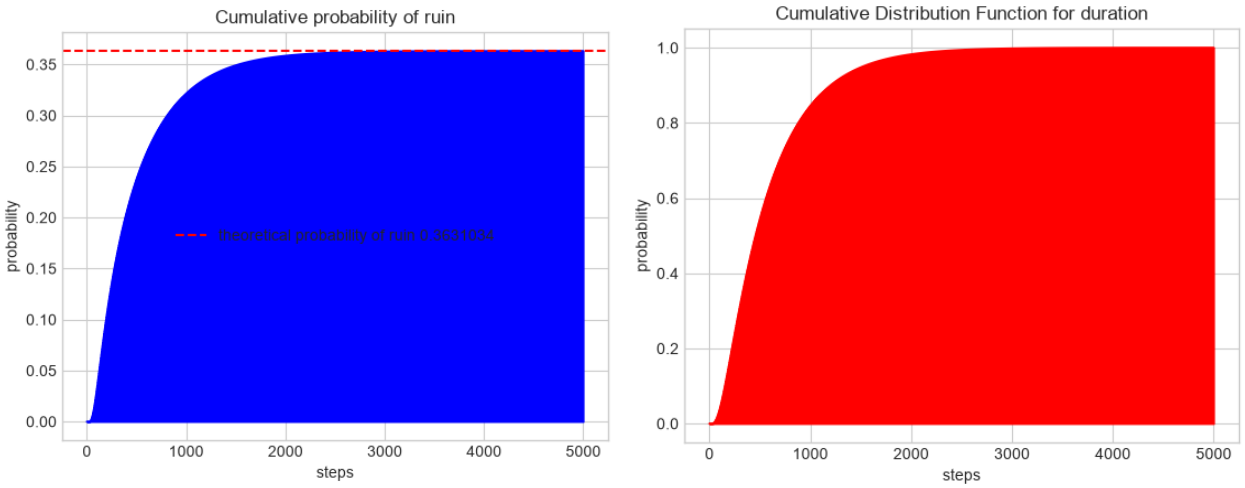


Figure 13: Figure of cumulative probability ruin and cdf of duration

The right plot in Figure (13) was computed by combining the cumulative probability of ruin and the cumulative probability of winning, which are the conditions required for the game to end. Following this logic the plot acts as the cumulative distribution function for the duration of the game, whilst being a much more elegant method than what we used to calculate the PMF previously.

5 Poisson processes

5.1 Defining a Poisson Process

Previously we have been working on stochastic processes in discrete space $S = \mathbb{Z}$ and discrete time steps. Poisson processes exist in a discrete space $S = \mathbb{N}$, however with continuous time. Picture a walk that can only move upwards but instead of just moving at each fixed step, there is a distribution deciding on when each step will be. This in fact is one of the ways we can define the Poisson process. More officially let $T_1, T_2, T_3, ...$ be i.i.d. random variables of time between each step with $T_i \sim \text{Exp}(\lambda)$ then

$$X(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq T_1, \\ 1 & \text{for } T_1 \leq t \leq T_1 + T_2, \\ 2 & \text{for } T_1 + T_2 \leq t \leq T_1 + T_2 + T_3, \\ \text{and so on.} \end{cases}$$

Figure (14) shows a visualization of what this process would look like if we had a rate of $\lambda = 0.5$ and max time of 10.

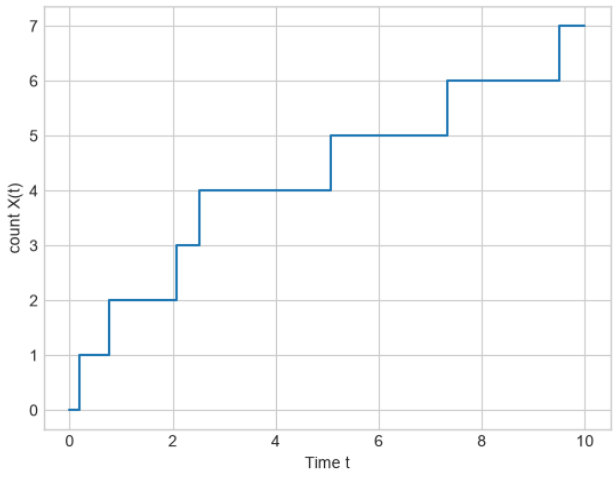


Figure 14: Visualization of Poisson process

We can also define a Poisson process using Poisson increments. That is a stochastic process $X(t)$ is a Poisson process if:

1. $X(0) = 0$
2. $X(t+s) - X(t) \sim Po(\lambda s)$ for all $s, t > 0$
3. $X(t_2) - X(t_1)$ and $X(t_4) - X(t_3)$ are independent for all $t_1 \leq t_2 \leq t_3 \leq t_4$

This definition and the first definition are equivalent, however using this definition we can easily work out the theoretical probability of a number of events happening in a given time. Whilst the first definition makes it easy to simulate a sample path whilst also being able to solve problems about waiting times.

5.2 Computational Verification

The left plot in Figure (15) was computed by generating $10,000$ paths of $\lambda = 0.5$ and max time of 10 where each bar represents the percentage of paths that ended with the final count at that position. The plot on the right is the PMF for a Poisson distribution with $\lambda = 5$. Notice that the plots are approximately the same this is due to the fact that $X(t) \sim Po(\lambda t)$. The expectation and variance of the final count was 4.975 and 4.896, respectively, and when we increase the number of paths to $100,000$ we get 5.012 and 5.017 converging closer to the theoretical value of 5.

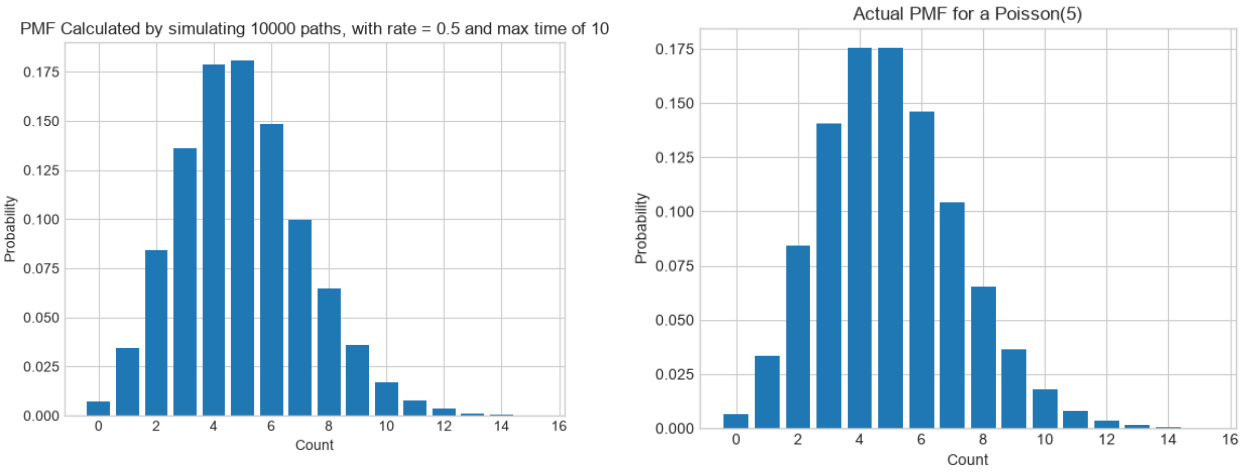


Figure 15: Figure comparing calculated pmf vs actual pmf

Let's also test the superposition property of Poisson processes, this being that if $X(t)$ and $Y(t)$ are both independent Poisson processes with rates λ and μ respectively then the process $Z(t)$ given by $Z(t) = X(t) + Y(t)$ is a Poisson process with rate $\lambda + \mu$. Let's pick $\lambda = 2$ and $\mu = 4$, we might expect that the mean and variance of generated values will converge to 6. Figure (16) shows this result closely.

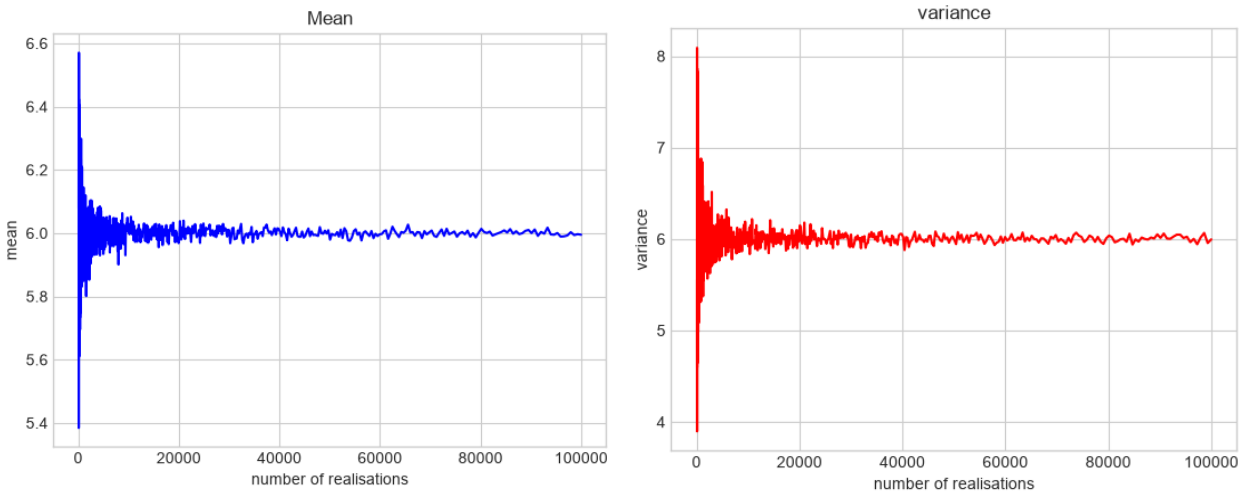


Figure 16: Expectation and variance as number of realizations increases

However we can better confirm this by looking at the PMFs. Figure (17) shows that when simulating $1,000,000$ realizations of $X(t) + Y(t)$ the difference in PMFs for theoretical $Z(t)$ and the simulated superposition become negligible.

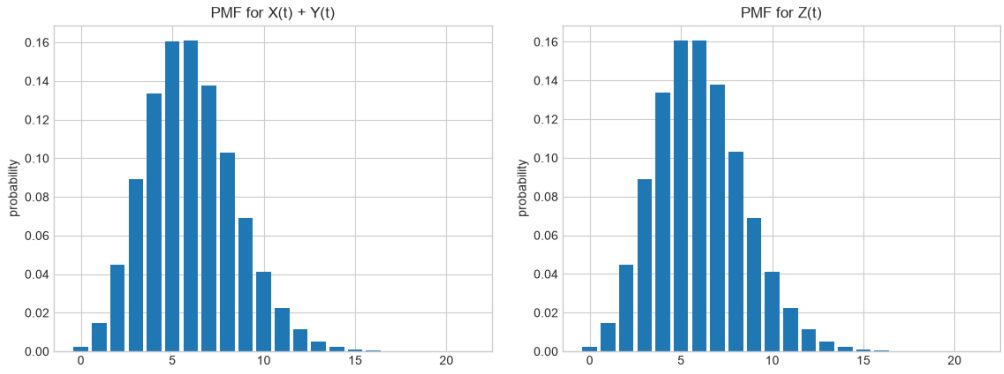


Figure 17: PMF for $X(t) + Y(t)$ and $Z(t)$

6 Brownian Motion

6.1 Definitions and Properties

Continuing with continuous time, let's introduce Brownian motion. Brownian motion is a continuous-time, continuous-state-space Markov process. A stochastic process $B(t)$ is defined as Brownian motion if it satisfies:

1. $B(0) = 0$ and $B(t)$ is a continuous function of t
2. independent increments: for any $0 < t_1 < t_2 < t_3 < t_4$ the increments

   $$B(t_2) - B(t_1) \text{ and } B(t_4) - B(t_3)$$

   are independent random variables.
3. stationary Gaussian increments: for every time t and time increment Δt,

   $$B(t + \Delta t) - B(t) \sim N(0, \sigma^2 \Delta t)$$

   where $\sigma > 0$ is the volatility parameter.

At this stage it is also important to introduce the Gaussian random walk, a discrete time continuous space Markov process. Earlier we defined a simple random walk as

$$X_n = X_0 + \sum_{i=1}^{n} Z_i, \quad \text{where} \quad Z_i = \begin{cases} 1 & \text{with probability } p, \\ -1 & \text{with probability } q. \end{cases}$$

. The Gaussian random walk follows a similar process however instead of using the step distribution Z_i with ± 1 for probabilities p and q, it uses independently distributed $\epsilon_i \sim N(0, \sigma^2)$. This gives

$$X_n = X_0 + \sum_{i=1}^{n} \epsilon_i, \quad \text{where} \quad \epsilon_i \sim N(0, \sigma^2).$$

Comparing the definitions of Brownian motion and Gaussian random walks, they both use the same incremental structure with independent and stationary Gaussian increments. They only differ in their time and state space with Gaussian random walks defined in discrete time $n \in \{0, 1, 2, \dots\}$ with fixed variance σ^2 and Brownian motion defined in continuous time with $t \geq 0$ and an increment variance $\sigma^2 \Delta t$ scaling with size of the time gap. Therefore discretizing Brownian motion onto a fixed time grid Δt gives a Gaussian random walk with step variance $\sigma^2 \Delta t$. This is the exact method we can use to simulate Brownian motion.

More generally, even non-Gaussian i.i.d. steps converge to Brownian motion in the limit, this is the content of Donsker's invariance principle which states, let $\{X_1, X_2, X_3, \dots\}$ be i.i.d. random variables with $E[X_i] = 0$ and $Var[X_i] = 1$. Define the random walk after n steps,

$$S_n = X_1 + X_2 + \cdots + X_n.$$

Normalise this walk by defining $B_n(t)$ for $t \geq 0$ as

$$B_n(t) = \frac{S_{[nt]}}{\sqrt{n}}$$

where $[nt]$ is the largest integer less than or equal to nt. Donsker's invariance principle states that B_n converges in distribution to a standard Brownian motion as $n \to \infty$.

Figure (18) shows this visually and as Δt decreases we see a clearer illustration of what Brownian motion looks like. When using Gaussian random walks at each point, there is no approximation error, they match the Brownian motion exactly, the only discrepancy is between the grid points but we know this vanishes as the grid is refined, so when using Gaussian random walks to simulate Brownian motion the key takeaway is that it's exact at the grid points but only an estimate between them.

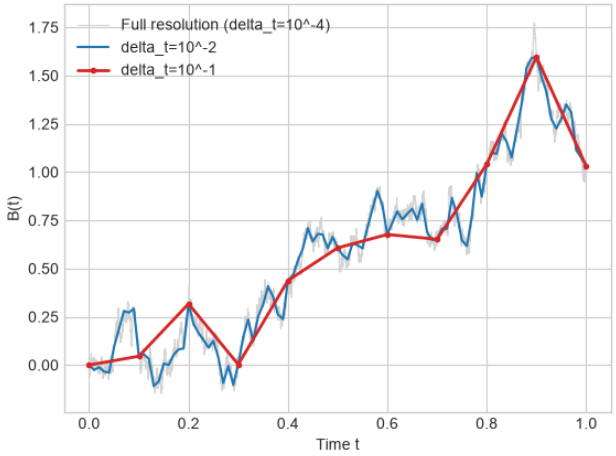


Figure 18: Image showing change in increment size Δt for the same Brownian motion

We can model probability by using the conditional distribution $Y = B(t)$ given $X = B(s)$,

$$[Y \mid X = x] \sim N(x, \sigma^2(t-s)) \qquad \text{for} \quad t > s$$

with a probability density function of

$$p(y, t \mid x) = \frac{1}{\sqrt{2\pi\sigma^2(t-s)}} e^{-\frac{1}{2}\frac{(y-x)^2}{\sigma^2(t-s)}}.$$

Also note that for any Brownian motion $B(t)$, $E[B(t)] = 0$ and $Var[B(t)] = \sigma^2 t$.

6.2 Simulating and Validating Properties

In the left plot of Figure (19) we observe a convergence for our estimates with an increase in sample paths n, importantly we also see that variance converges to a straight line with gradient $\sigma^2 = 1$. We can see that for all time t the expectation of $B(t)$ is 0.

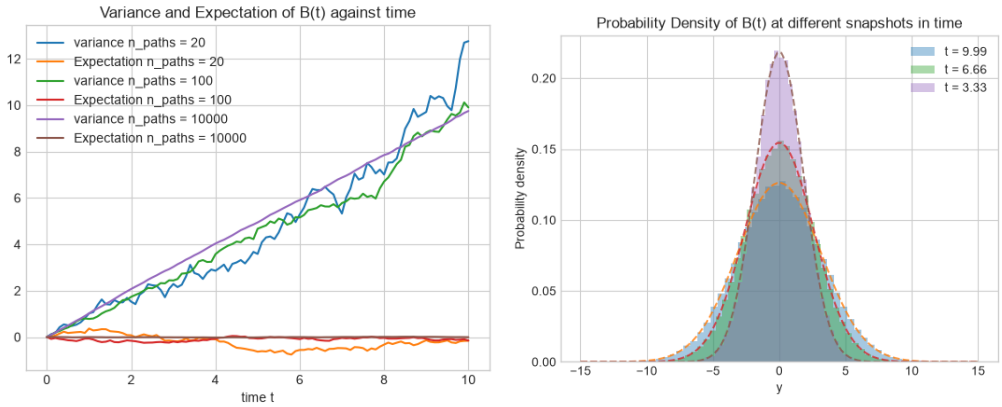


Figure 19: Figure showing empirical expectation, variance and probability density over time

The right side plot was obtained by computing the histogram of $100,000$ paths at different snapshots in time, with the dashed lines added corresponding with the theoretical PDF. We can see that for the earlier snapshot at time $t = 3.33$, that the spread is much smaller than the other 2, following a normal distribution with mean of 0 and standard deviation of $\sqrt{3.33}$. For $t = 6.66$ and $t = 9.99$, we can see a gradual increase in spread, let's investigate this further by looking at how the theoretical PDF itself evolves over time. Figure (20) shows a surface plot of density and spread y against time t. Starting from time $t = 2$ we see the probability density is concentrated but as t increases we see a bigger spread and the concentrated probability diffuses outwards. The width of the bump at time t scales like standard

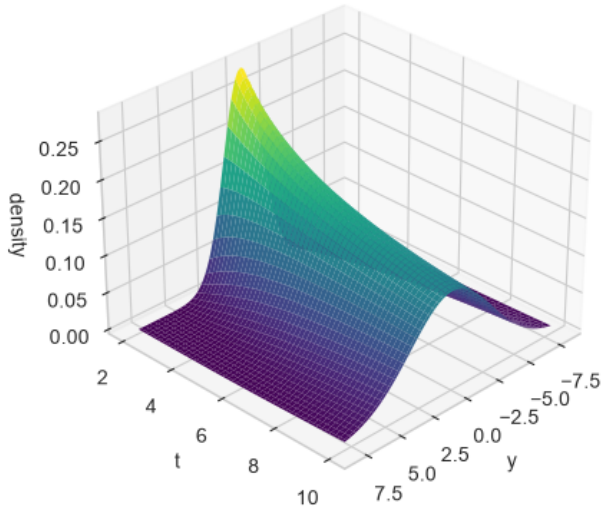


Figure 20: Figure showing diffusion of probability over time

deviation $\sigma\sqrt{t}$ so the spreading actually increases over time. This structure happens for a reason, which will be explained now.

We established earlier $B(t) \sim N(0, \sigma^2 t)$ with PDF $p(y, t \mid x)$, now let's introduce the diffusion equation that the PDF satisfies:

$$\frac{\partial p}{\partial t} = \frac{1}{2}\sigma^2 \frac{\partial^2 p}{\partial y^2}.$$

At the peak $y = 0$ the shape of the curve is locally concave so $\frac{\partial^2 p}{\partial y^2} < 0$ and therefore $\frac{\partial p}{\partial t} < 0$, so the peak is falling which is exactly what we see in the plot. Similarly at the tails which are convex we have $\frac{\partial^2 p}{\partial y^2} > 0$ so $\frac{\partial p}{\partial t} > 0$, telling us that the ends are rising.

It is worth noting that the diffusion equation isn't really necessary for standard Brownian motion as solving the PDE would give us the same PDF as defining Brownian motion by gaussian increments. The diffusion equation is mostly useful for its generality as later it could be used for solving more complex equations.

6.3 Two Boundary Hitting Times for Brownian Motion

Let's consider a standard Brownian motion with two boundaries, one at 0 and the other at m, where we want to know the expected time before hitting a boundary and the probability of hitting each boundary. In essence, this is a variation of the Gambler's ruin problem, however in continuous time and state space.

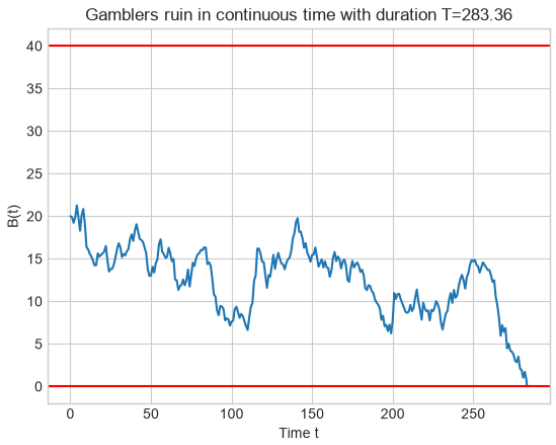


Figure 21: Gambler's ruin in continuous time

Figure (21) shows a simulated realization of a Brownian motion in a gambler's ruin problem with start position $a = 20$ and barriers at $m = 40$ and 0, the hitting time T was estimated by linearly interpolating the simulated path between the last point inside the barriers and the first point outside them.

Like before let's first derive formula for probability of ruin and expected duration, before this let's introduce martingales.

Martingales are essentially just stochastic processes where the expected future position is the position it is in currently,

$$E[X_{t+n} \mid X_0, \ldots, X_t] = X_t.$$

Brownian motion is also a martingale, since its increments have zero conditional expectation.

We now consider a Brownian motion starting at a, so that $B_0 = a$, and fix two boundaries at 0 and m. Let's define the stopping time

$$T = \inf\{t \geq 0 : B_t = 0 \text{ or } B_t = m\},$$

the first time the Brownian motion hits either boundary. We want

$$p(a) = \mathbb{P}(B_T = 0 \mid B_0 = a),$$

the probability of hitting 0 before m. Since T is the first hitting time of either boundary of the bounded interval $[0, m]$, the optional stopping theorem can be applied to the martingale B_t. Hence

$$E[B_T] = E[B_0] = a.$$

Using the fact that there can only be two outcomes, $B_T = 0$ with probability $p(a)$ and $B_T = m$ with probability $1 - p(a)$, we can solve for $p(a)$,

$$\begin{aligned} E[B_T] = 0 \times p(a) + m \times (1 - p(a)) &= a \\ m(1 - p(a)) &= a \\ p(a) = 1 - \frac{a}{m} &= \frac{m - a}{m}. \end{aligned}$$

To solve for expected duration we need to use a clever trick since $E[B_T]$ alone can't give us $E[T]$ since B_t carries no information on time elapsed, so in order to solve $E[T]$ we need a martingale that explicitly involves t, in this case we will use the martingale $X_t = B_t^2 - t$. To prove X_t is in fact a martingale we need to make sure

$$E[B_{t+n}^2 - (t+n) \mid B_0, \ldots, B_t] = B_t^2 - t$$

is satisfied. Let

$$B_{t+n} = B_t + \Delta, \quad \text{where} \quad \Delta = B_{t+n} - B_t \sim N(0, n) \text{ is independent of } B_0, \ldots, B_t.$$

Then,

$$B_{t+n}^2 = B_t^2 + 2B_t\Delta + \Delta^2$$

and by taking expectations with $E[\Delta] = 0$ and $E[\Delta^2] = n$ we get,

$$E[B_{t+n}^2 \mid B_0, \ldots, B_t] = B_t^2 + 2B_t \cdot E[\Delta] + E[\Delta^2] = B_t^2 + n$$

so,

$$E[B_{t+n}^2 - (t+n) \mid B_0, \ldots, B_t] = B_t^2 + n - (t+n) = B_t^2 - t.$$

This confirms that $X_t = B_t^2 - t$ satisfies the martingale property. We can again use the optional stopping theorem,

$$\begin{aligned} &E[X_T] = E[X_0], \\ &E[B_T^2 - T] = E[B_0^2 - 0] = a^2 \\ &E[B_T^2] - E[T] = a^2. \end{aligned}$$

Using the ruin probability from before we can compute $E[B_T^2]$ as

$$E[B_T^2] = 0^2 \times p(a) + m^2 \times (1 - p(a)) = m^2 \times \frac{a}{m} = am$$

and finally we can substitute into $E[B_T^2] - E[T] = a^2$ to solve for $E[T]$ where we obtain

$$E[T] = am - a^2 = a(m - a).$$

From the formulas we have derived there is a linear relationship between probability of ruin and start position and a quadratic relation between expected duration and start position. Before verifying this computationally let's first investigate how the accuracy of these formulas changes with sample size.

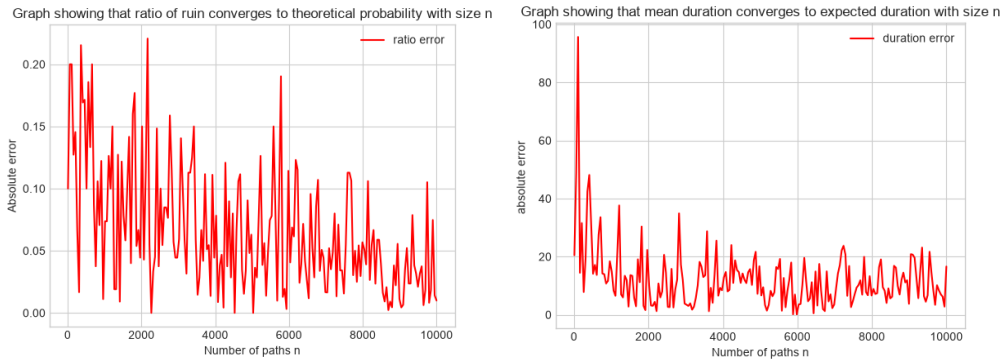


Figure 22: Figure showing the convergence of simulated values

As seen in figure (22) the computationally simulated values become closer to the theoretical with increasing sample size n, they do so in a way similar to what we observed in the classical gambler's ruin investigation, that is we see a decrease in absolute error roughly proportional to $1/\sqrt{n}$. Fixing the boundary position $m = 50$ and sample size $n = 1000$, we can verify the relation of our formulas with start position by computing probability of ruin and expected duration with change in a. Figure (23) shows these relationships from the theoretical formulas, with probability of ruin decreasing linearly with the starting position and expected duration following a negative quadratic relationship.

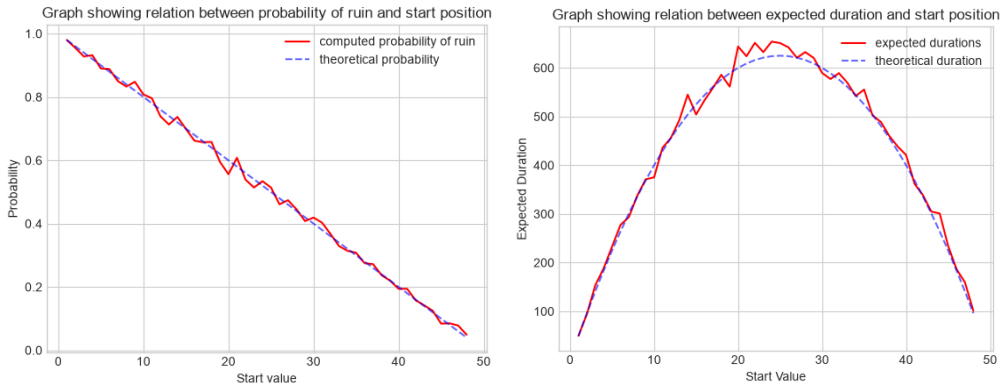


Figure 23: Figure showing the relation between Probability of Ruin / Expected Duration and start position

This analysis on gambler's ruin has only considered the case of a standard Brownian motion without any drift, which is what allows us to use the martingale property directly to solve for probability of ruin and expected duration. Brownian motion with drift is an obvious next step, one that I would like to investigate as further work or an extension to this project. In order to solve this, we could construct an appropriate exponential martingale using Itô's formula.

7 Future Research

Brownian motion with drift

The continuous gambler's ruin analysis considered only standard Brownian motion, which allowed the martingale property to be used directly. Adding a drift removes this, but an exponential martingale constructed with Itô's formula should give the probability of ruin and expected duration. This would be the continuous analogue of the biased case $p \neq 0.5$ studied in the discrete setting.

Convergence from discrete to continuous

Donsker's invariance principle suggests that the discrete gambler's ruin, appropriately rescaled, converges to the Brownian version. I would like to measure how quickly the discrete ruin probability and expected duration approach the continuous formulas as the step size decreases.

Applications to finance

Many of the processes studied here have direct financial counterparts, and this was the original motivation for the project. Gambler's ruin is closely related to the risk of an investor or firm reaching zero capital before hitting a target. The two-boundary hitting time problem for Brownian motion is the basis for pricing barrier options, where a payoff depends on whether an asset price crosses a level. Extending Brownian motion with drift to geometric Brownian motion would give the standard model of asset prices underlying the Black–Scholes framework, and the same Monte Carlo methods used here could be used to price options by simulation. Poisson processes could be used to model jumps in asset prices or the arrival of insurance claims, and a Markov chain with an absorbing default state could model credit rating transitions and the probability of default over time.

Variance reduction

The simulations in this report showed that Monte Carlo error falls only as $1/\sqrt{n}$, which becomes costly for the large path counts needed in financial applications. Techniques such as antithetic variates and importance sampling could achieve the same accuracy with far fewer paths, and I could test how much they improve the ruin probability estimates.

Exact hitting time distributions

For the discrete game, the transition matrix gave an exact duration distribution. A natural extension is to find the corresponding distribution of the hitting time for Brownian motion between two barriers, for example by solving the diffusion equation with absorbing boundary conditions.